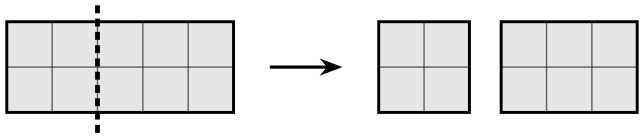
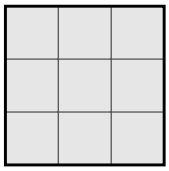
A break splits one piece into two pieces along a straight line between squares. Break one piece at a time, and never stack pieces. In the game “last break wins,” two players take turns making one break, and whoever makes the last break wins.

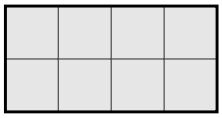


Problem 1: Play “last break wins” on the 3 by 3 bar and the 2 by 4 bar. Play each bar twice, switching who goes first, and circle whether the 1st or 2nd player won. Can the losing player ever win by breaking in a different way?



3 by 3

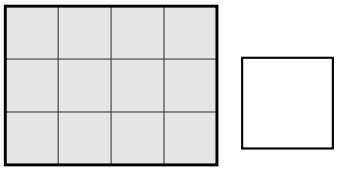
Game 1: 1st 2nd Game 2: 1st 2nd



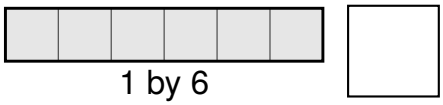
2 by 4

Game 1: 1st 2nd Game 2: 1st 2nd

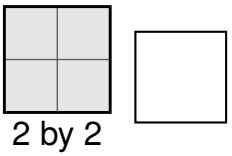
Problem 2: Find the fewest breaks that split the 3 by 4 bar into single squares. Then find the fewest for the 1 by 6 bar and the 2 by 2 bar.



3 by 4

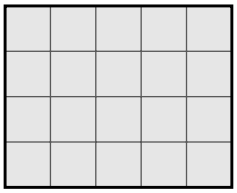


1 by 6



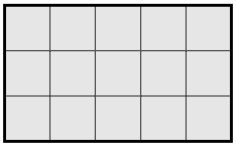
2 by 2

Problem 3: Decide who would win “last break wins” on each of these bars without breaking them, and explain how you know.



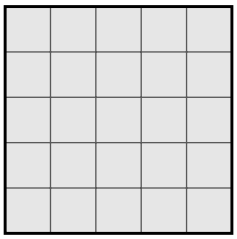
4 by 5

1st 2nd



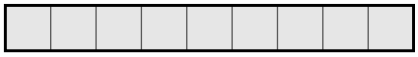
3 by 5

1st 2nd



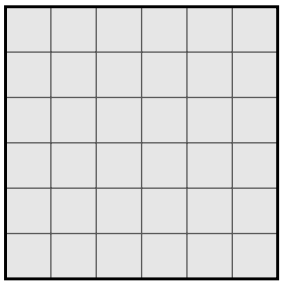
5 by 5

1st 2nd



1 by 9

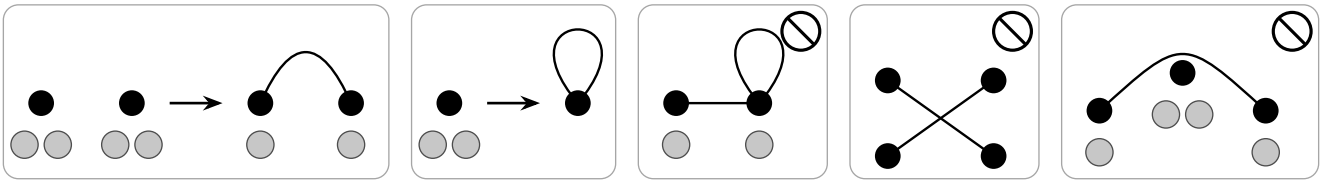
1st 2nd



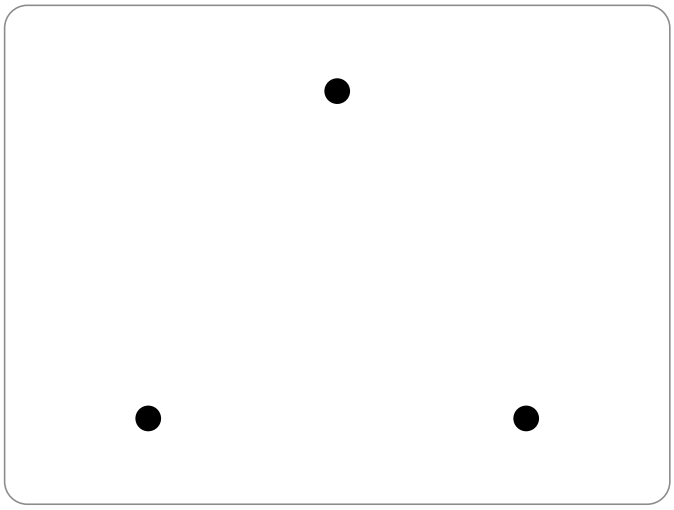
6 by 6

1st 2nd

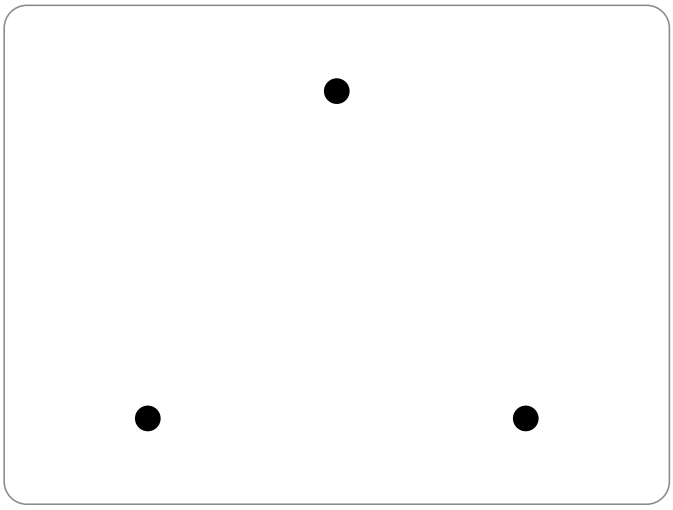
In 2-Sprigs, each dot starts with two tokens beside it. Players take turns drawing a line from a dot to another dot or back to the same dot. A line may bend, but it may not cross itself or another line, or touch any other dot. Each end of a line uses up one token at its dot, so a loop needs a dot with two tokens. After each line, the other player takes away the tokens it used. Whoever draws the last line wins.



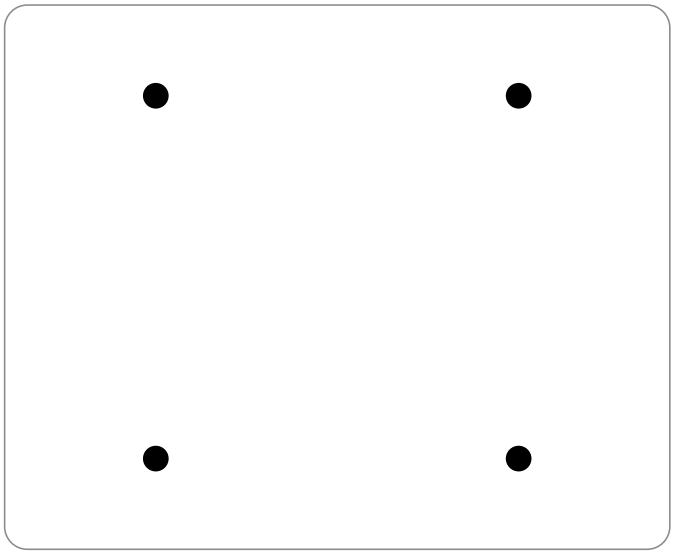
Problem 4: Play 2-Sprigs on these boards. Write how many moves each game lasted, and circle who won.



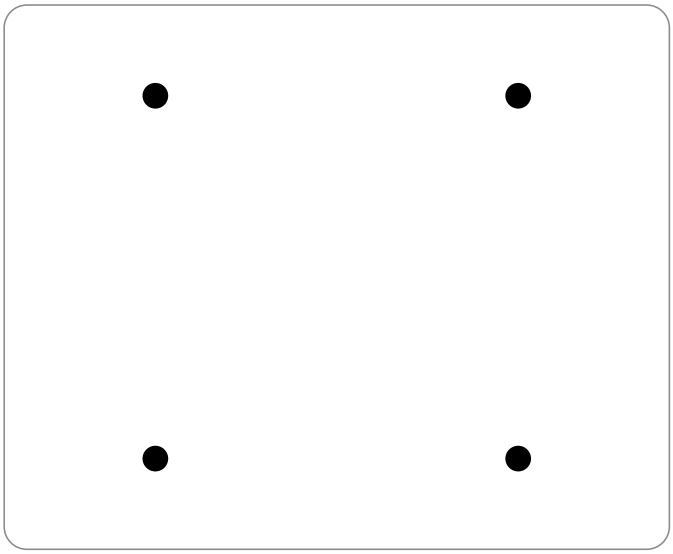
Moves: _____ Winner: 1st 2nd



Moves: _____ Winner: 1st 2nd

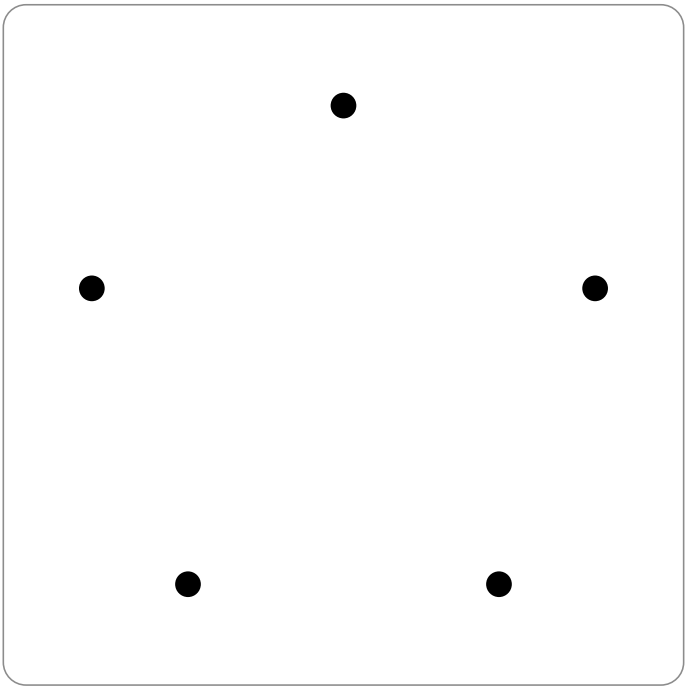


Moves: _____ Winner: 1st 2nd



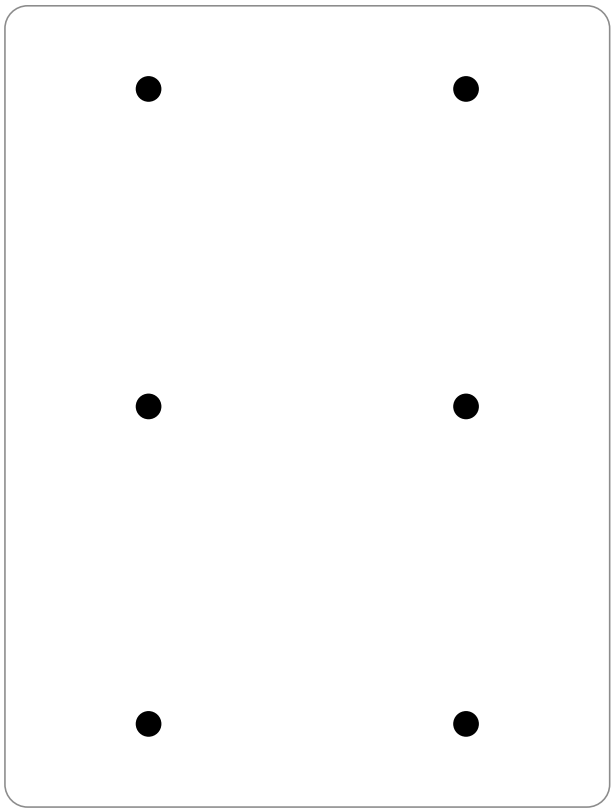
Moves: _____ Winner: 1st 2nd

Problem 5: Before you play 2-Sprigs on each board, decide who will win. Then play to check. Who would win 2-Sprigs with 10 dots? With 25 dots? Explain how you know.



Prediction: 1st 2nd

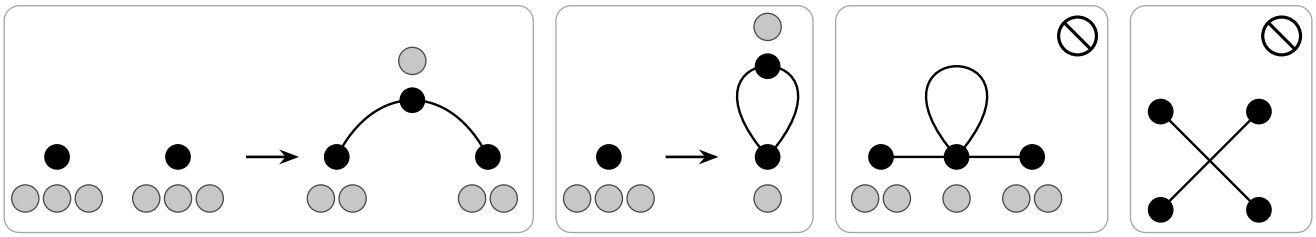
Moves: _____ Winner: 1st 2nd



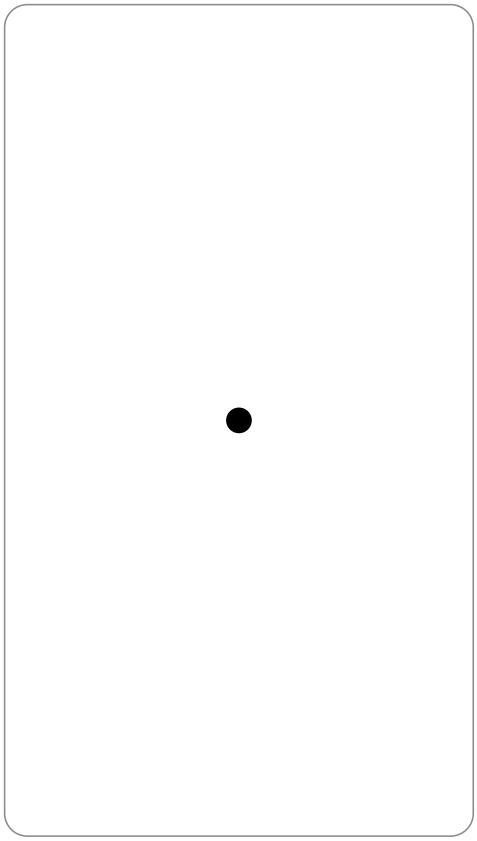
Prediction: 1st 2nd

Moves: _____ Winner: 1st 2nd

In Sprouts, each spot starts with three tokens. Players take turns drawing a line from a spot to another spot or back to the same spot. A line may bend, but it may not cross itself or another line, or touch any other spot. Each end of a line uses up one token at its spot, so a loop needs a spot with two tokens. After each line, the other player takes away the tokens it used, and the player who drew it adds a new spot on the line and gives it one token. Whoever makes the last move wins.

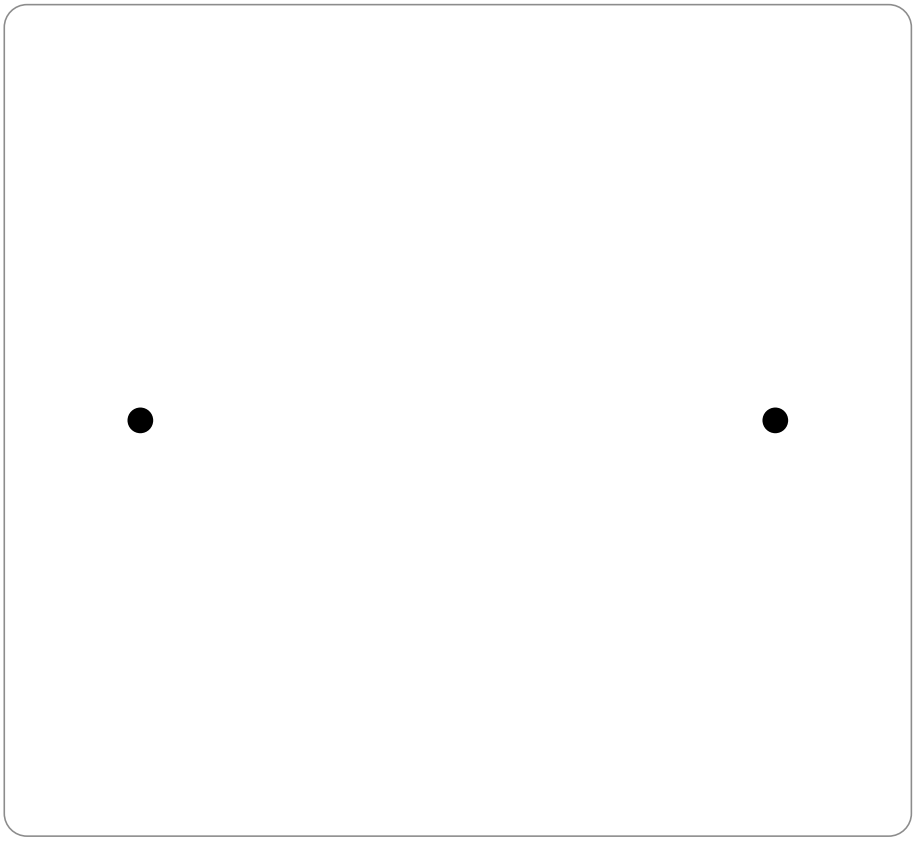


Problem 6: Play Sprouts starting with 1 spot, and then starting with 2 spots. Write how many moves each game lasted, and circle who won.



Moves: _____

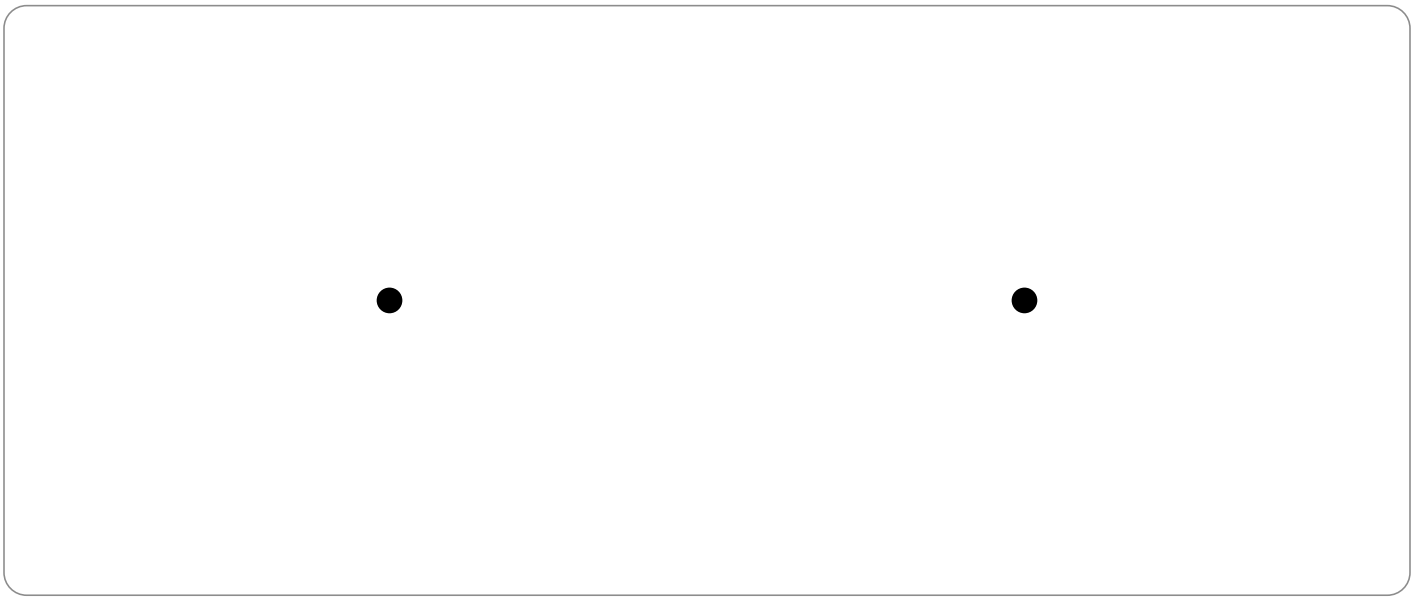
Winner: 1st 2nd



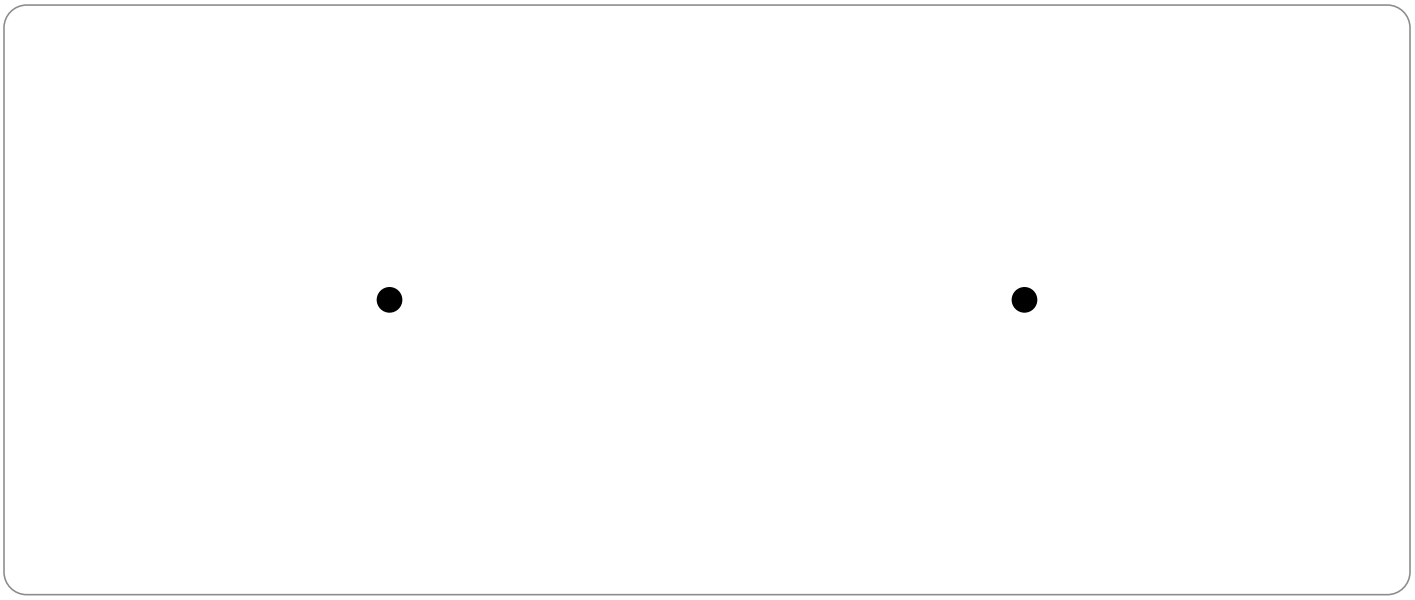
Moves: _____

Winner: 1st 2nd

Problem 7: Work together to make one 2-spot game of Sprouts last as many moves as you can, and another end after as few moves as you can. Can the players' choices change who wins Sprouts? Explain how you know.



Moves: _____



Moves: _____

Problem 8: What is the largest number of moves a game of Sprouts that starts with 3 spots can last? Explain how you know.

